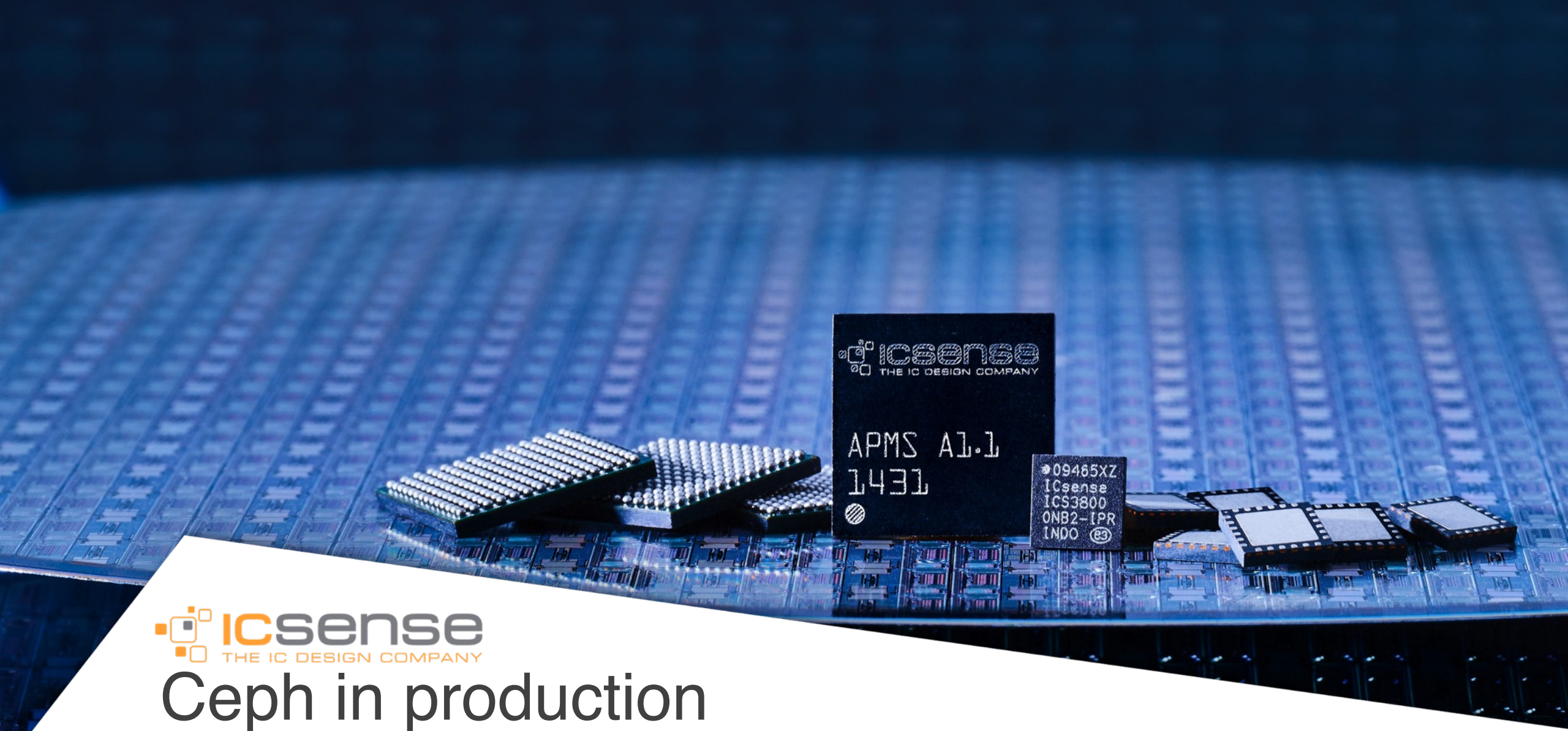


WELCOME

Wannes Smet



14th October 2026, Zurich



icsense
THE IC DESIGN COMPANY

Ceph in production

Things we swore we'd never do twice

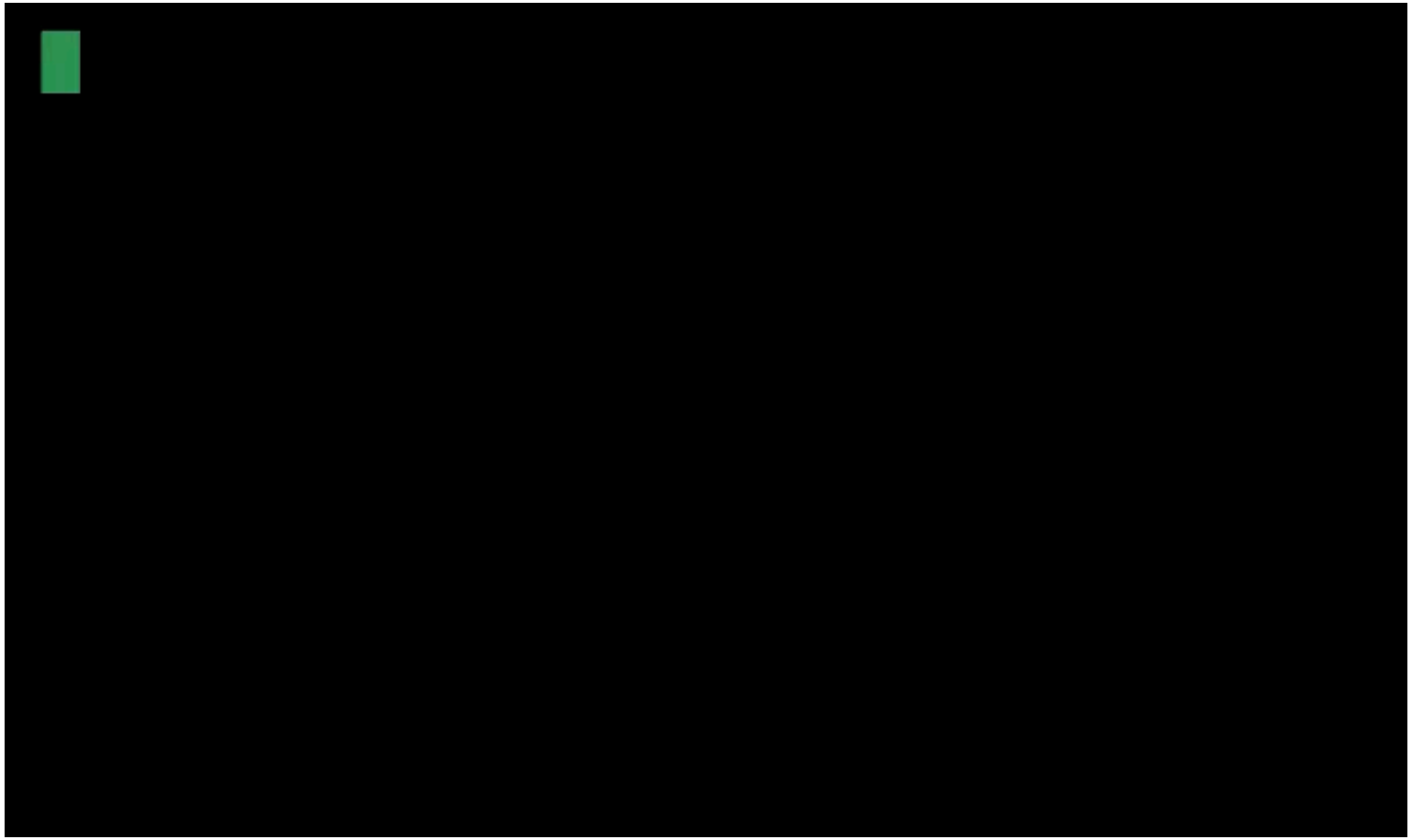
October 2026 | Wannes Smet

Agenda

- Intro & Framing
- Cluster Context
- War stories:
  - Because losing quorum once is not enough.
  - The balancer’s useful. Who could’ve known!?
  - CephFS going stale on node entering maintenance mode
  - Big crushmap changes & misplaced objects.
  - Changelogs
- Key takeaways
- References
- Q&A

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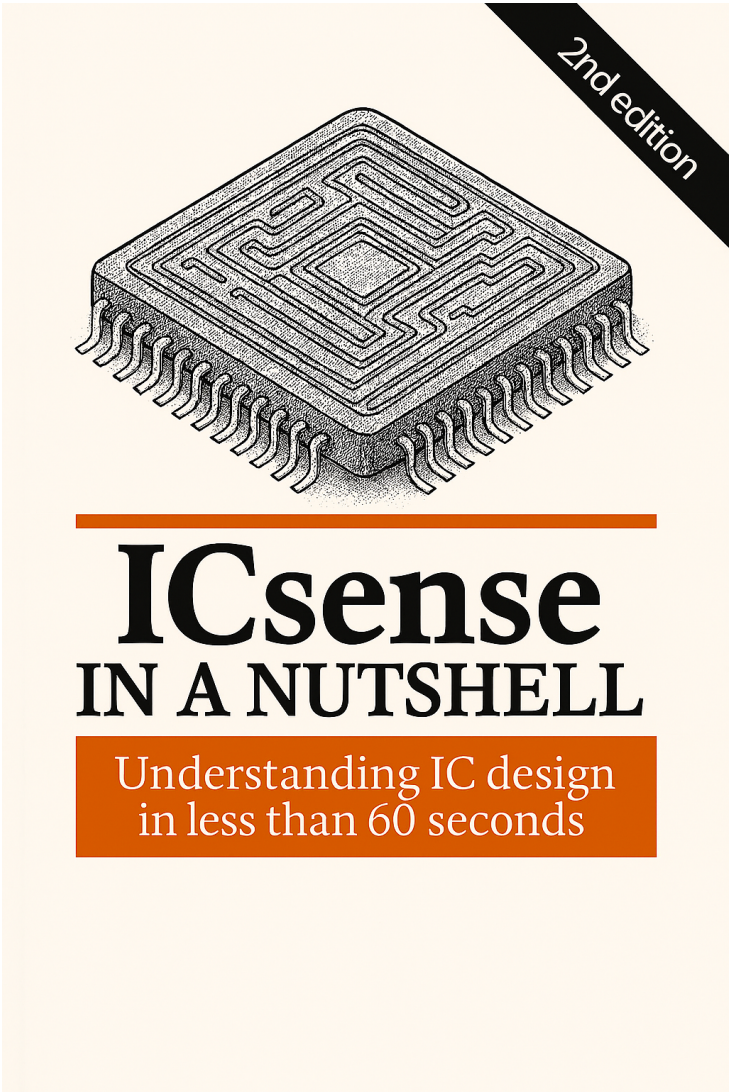
Intro & Framing



What do we do?

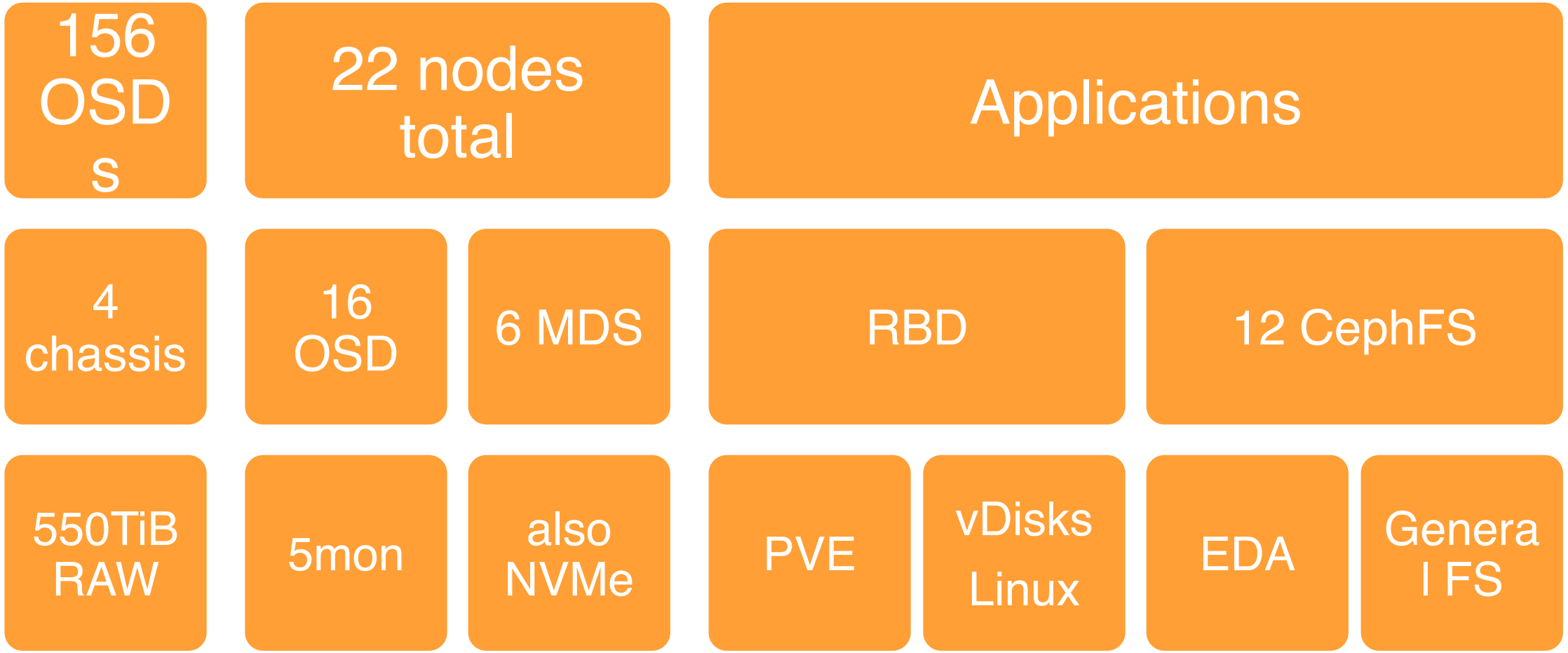
- IC / ASIC design company
- Extensive chip simulation for design verification
- Goal: eliminate design errors before fabrication
- ASICs don’t run software → no post-silicon fixes possible
- Path from idea to tape-out is costly.
- Fixes ≈ 6 months delay
- First-Time-Right and automation is key for us.

- We serve medical, automotive, aerospace, ... markets
- Errors are not acceptable



Intro & Framing

Our cluster at a glance



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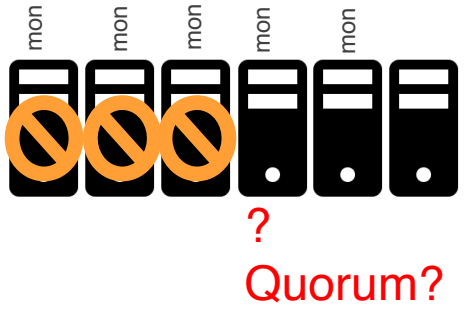
Because losing quorum once is not enough: 1st Round

February ~ March 2025, starting out with Ceph and feeling out the waters.

- PoC cluster: 6 nodes, 12 OSDs, 5 mons, low fill rate, only data produced by **`rados bench`**.
- The plan: Ceph is Self healing & scale out/down, let's try removing hosts until 2 are left.

- Pulled the first node.
  - Short lock-up (6s), 10 minutes later, recovery starts, Great stuff!
- Pulled second node
  - Short lock-up (6s), 10 minutes later, recovery starts, Great stuff!
- Pulled the third node

- Turns out, quorum doesn't scale as nicely as data flows around in the cluster during recoveries.
- Business impact: none. Luckily, I was just messing around with a PoC cluster.



Biggest lesson:

- I experienced first-hand what loss of quorum looks and feels like
- This new insight became very valuable in the next slide.

Because losing quorum once is not enough: 2nd Round

The plan:

- September 1st 2025. At exactly 07:00AM in the morning.
- Meanwhile the cluster is running some production workloads.
- Writing a support email and spotted weird ownership (`167:167`) in `/var/lib/ceph`. Looks wrong
- I use my google-fu and sure enough, RedHat site tells me it's a bug and needs to be chowned to `ceph:ceph`
- `for cephnode in $(ceph orch host ls | ... ); do ssh $cephnode chown -R ceph:ceph /var/lib/ceph; done`.
- <ENTER>, let the games begin.

The result:

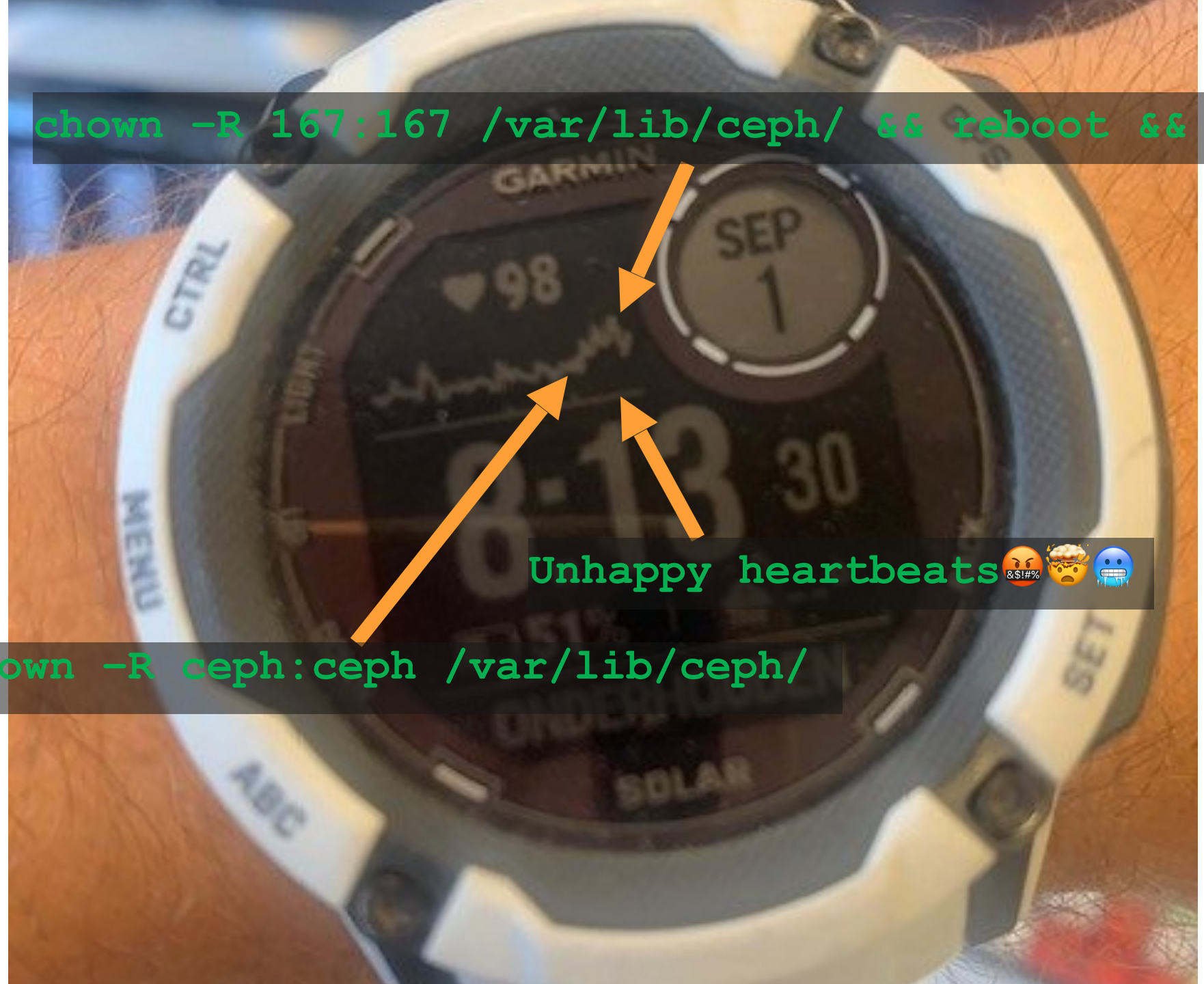
- Each daemon in our cluster crashed. Including monitor daemons. So I lost quorum with a one-liner.
- A massive déjà vu to 6 months prior. A bit of luck: I instantly knew: "lost quorum!!"
- Ceph operator PTSD because:
  - some production VMs running on RBD images
  - Support contract was "almost" in place.
  - Being the "Ceph guy" is all fun and games, ... until stuff like this happens.

Misconceptions:

- I was changing ownership of folders of running podman containers. Don't do that!
- Don't apply RedHat documentation for package installed Ceph to Debian + cephadm/podman!
- My `for` loop should have included "Press Enter to continue to next node" while checking the status of the cluster.

Fix:

- `chown -R 167:167 /var/lib/ceph/ && reboot # or podman restart`
- `ceph orch daemon redeploy ...`


chown -R 167:167 /var/lib/ceph/ && reboot && 🤞
Unhappy heartbeats🥵🤯🥶
chown -R ceph:ceph /var/lib/ceph/

The balancer ’s useful. Who could have known!?

Situation:

- Dec 2025: a warning email from our cluster: osd.25 & 6 pools **`nearfull`**
- Really odd and unexpected because our cluster wide fill rate was around 60%~65%.

```
HEALTH_WARN

--- New ---
[WARN] OSD_NEARFULL: 1 nearfull osd(s)
       osd.25 is near full
[WARN] POOL_NEARFULL: 6 pool(s) nearfull
       pool '.mgr' is nearfull
       pool 'pve' is nearfull
       pool 'cephfs.icsense.meta' is nearfull
       pool 'cephfs.icsense.data' is nearfull
       pool 'homedirs_data' is nearfull
       pool 'homedirs_metadata' is nearfull
```

Actions:

- Delete test data: back to **`HEALTH_OK`**
- Filed a support ticked which went more less like:
  - Support: **`ceph balancer status`**?
  - Me: I disabled it ...

My reasons:

- I wasn’t aware of what the balancer really does.
- Because I disabled it during the Ceph training. (Blame’s on me 100%(*))
- I tend to dislike “black box algorithms” causing unexpected IO in my cluster, I simply disabled the balancer.

The fix:

- **`ceph balancer on # and check its settings`**
- I kept an eye on **`watch -n60 -d ceph osd df`**

(*) The training material literally says this:

> ... The balancer is great for a production environment, but it does randomize our results with regard to the training ...

CephFS shares going stale

Situation:

- Entering dedicated MDS nodes causes our bigges CephFS filesystem to go stale.
- Mounts on almost all clients went stale, manual `umount -lf /mnt/simulres && mount -a` required on all clients.

Combination of things wrong:

- We're stuck at RHEL8, kernel 4.18. The mounts going stale is highly likely due to that.
- #MDS daemons/host + 100M file systems is RAM intensive → Linux kills it OOM during fail-over

Fix ~ workarounds:

- Reshuffle RAM from OSD/MON nodes to MDS nodes
- Either wait till we can upgrade to RHEL9? Or try to compile our own kernel based on 5.15LTS?

Not a fix in our case:

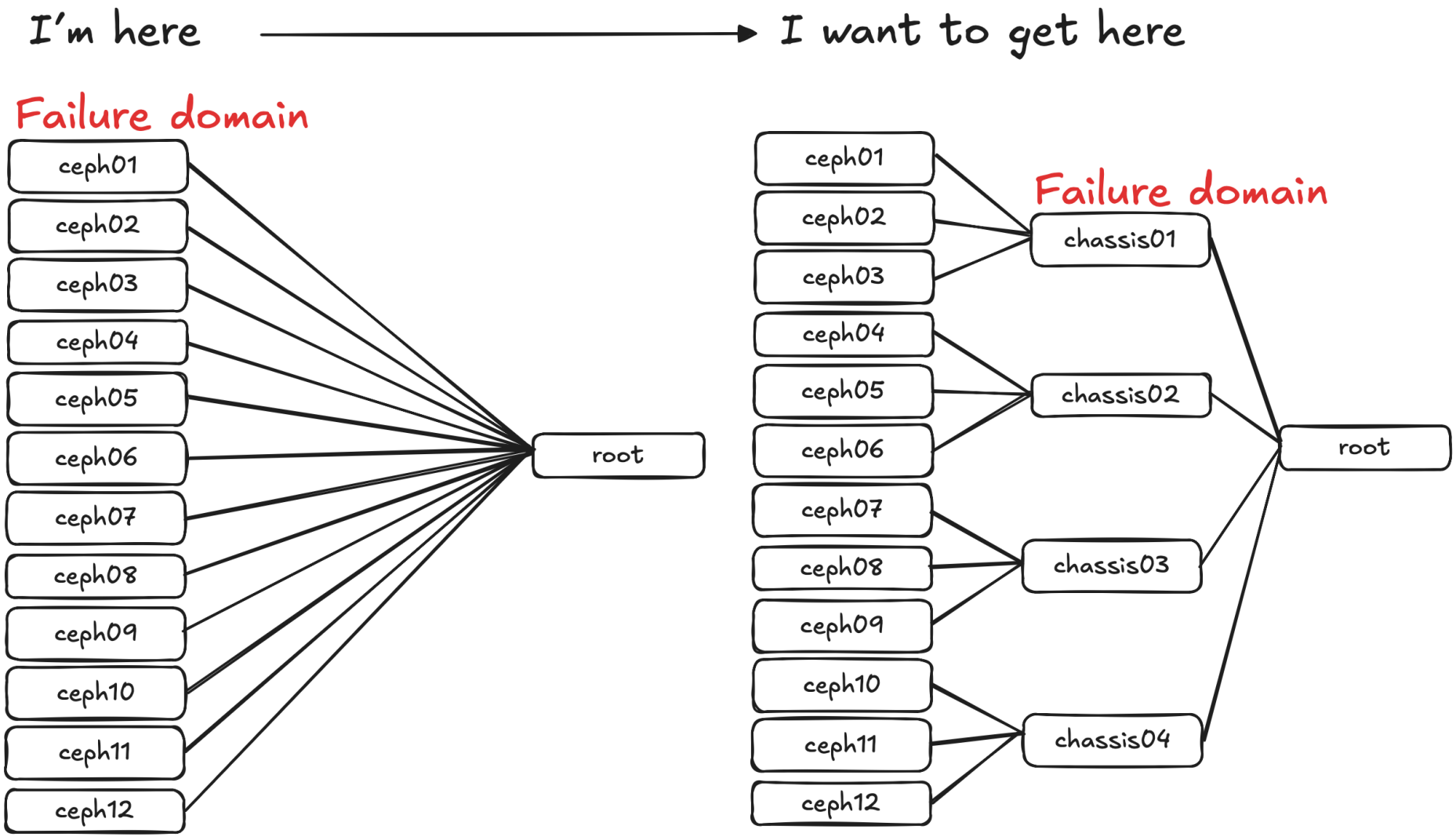
- We tried ceph-fuse
  - ceph-fuse SEGFAULT on 2 hosts on the first day (18.2.2, possibly fixed in 18.2.5 but didn't persue because:
  - Ceph-fuse is way too slow on our workload scenario.

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Crushmap changes & misplaced objects

Situation:

- I wanted the crushmap better reflectour physical setup to survive a loss of a chassis.



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Crushmap changes & misplaced objects

Situation:

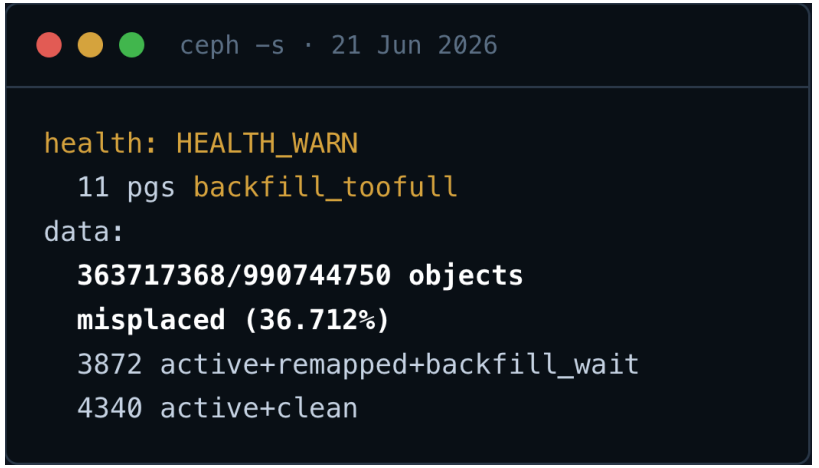
- I needed to bump up the failure domain from host to chassis
- Is going to induce PGs flowing around in the cluster.
- I had a short maintenance window so, in the end I decided to write a script that moves host from root=default to chassis=enclosure0x, all in one go <30s.

The result:

- A gradual rise of PGs in backfill_toofull
- While setting osd_recovery_max_active to 1 stopped the rise of PGs in backfill_toofull, I'm not 100% convinced this actually stopped it.

What I wouldn't do twice:

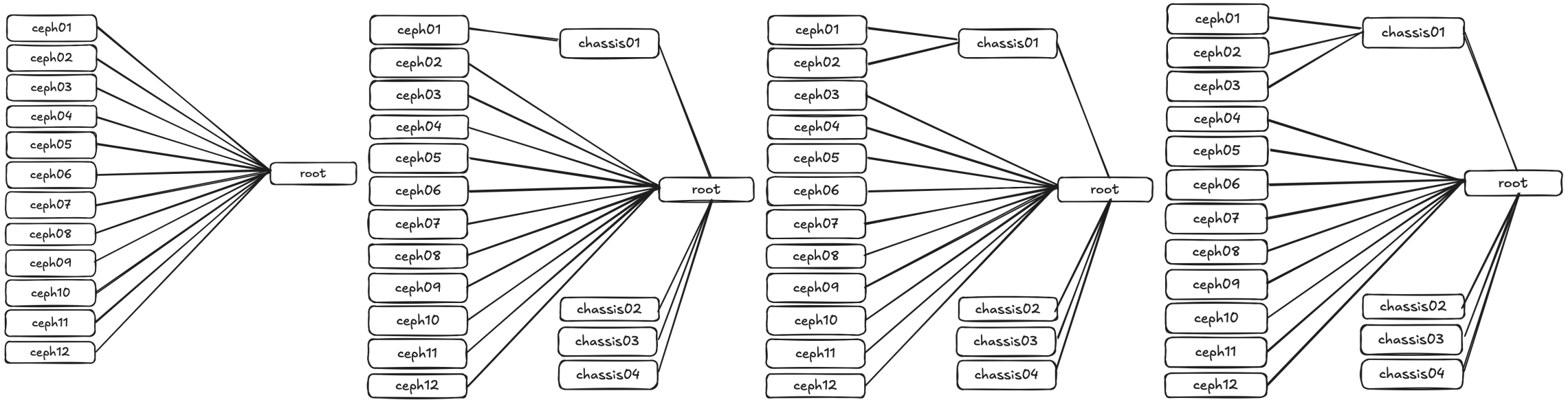
- Rethink maintenance windows and don't force big changes on the cluster in a short amount of time.
- Attempts to reproduce in a ceph labs weren't overly convincing, but I believe host by host to be gentler.



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Crushmap changes, 36.7% misplaced objects

What I should have done: step by step.



1. Move one node to chassis
2. Wait until recovery finishes

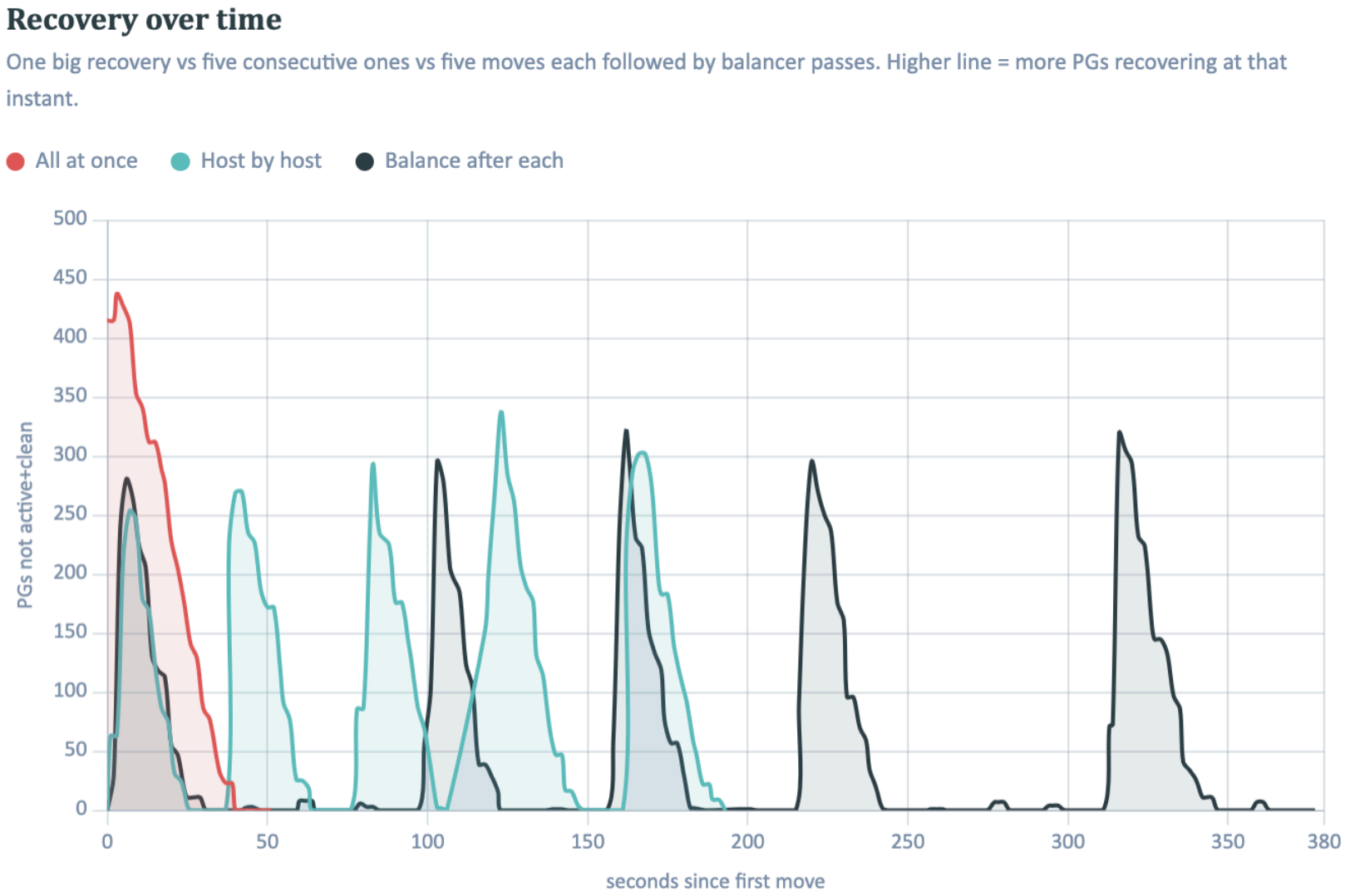
1. Move one node to chassis
2. Wait until recovery finishes

1. Move one node to chassis
2. Wait until recovery finishes

Repeat the process for the remaining 9 nodes
Every step wait for HEALTH_OK and no more data movement.

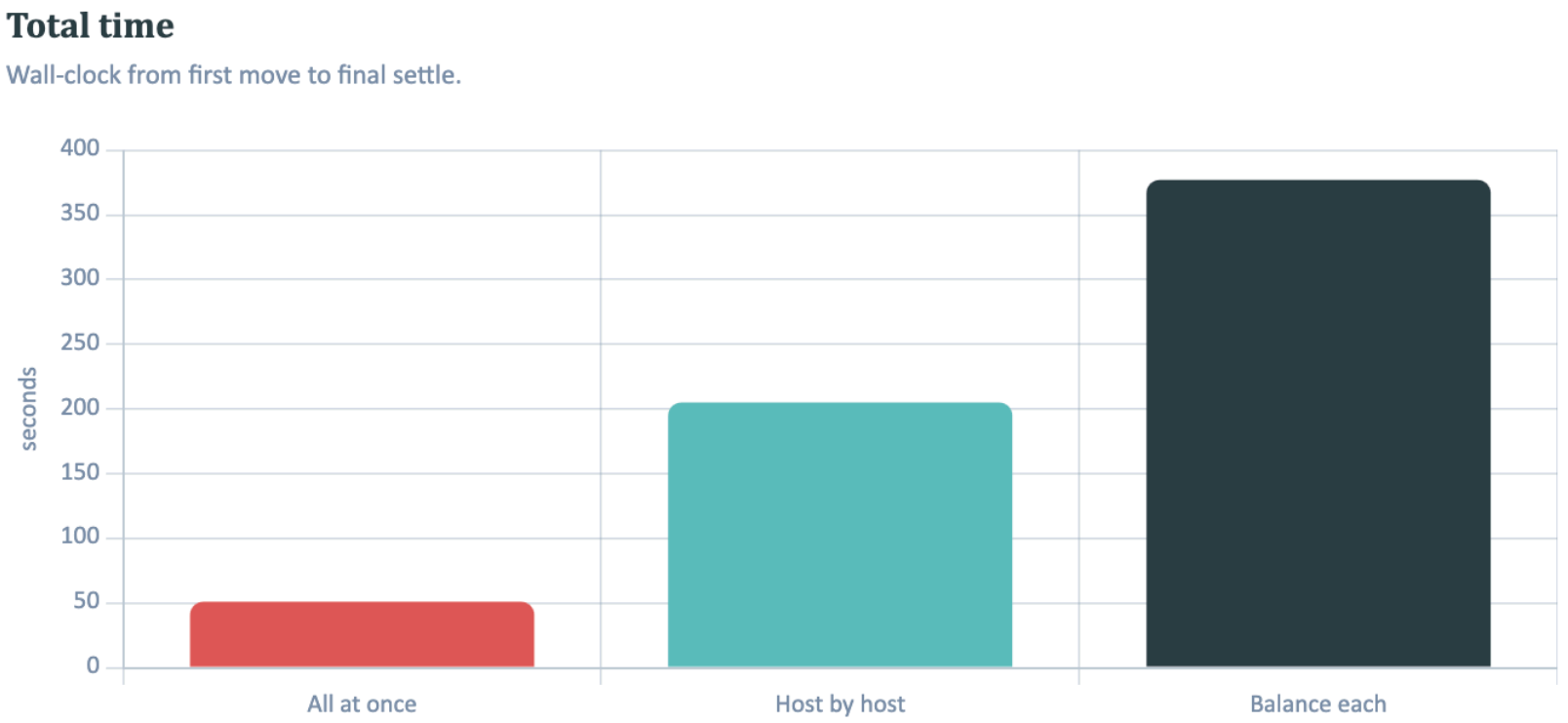
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Crushmap changes big-bang vs gradual & +=balancer

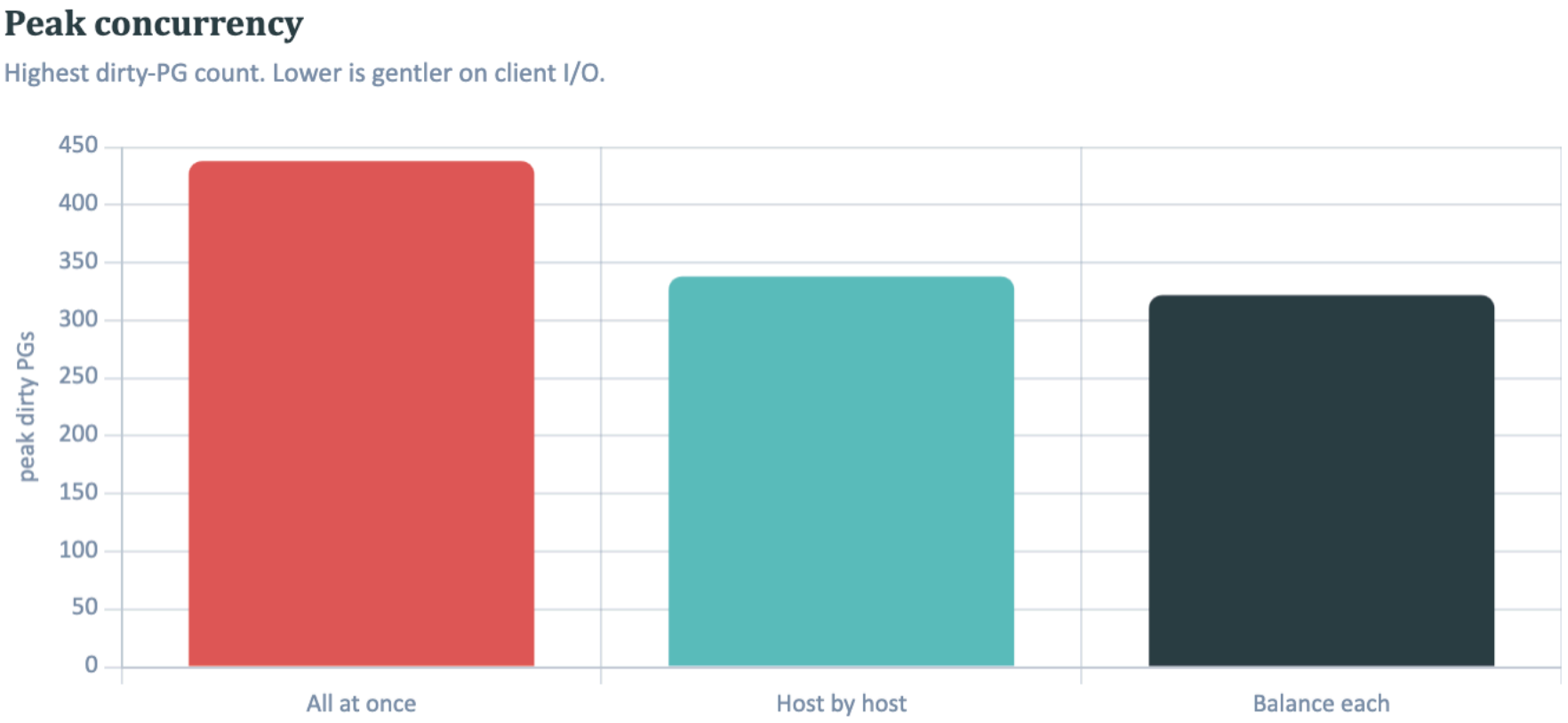


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Crushmap changes big-bang vs gradual & +=balancer

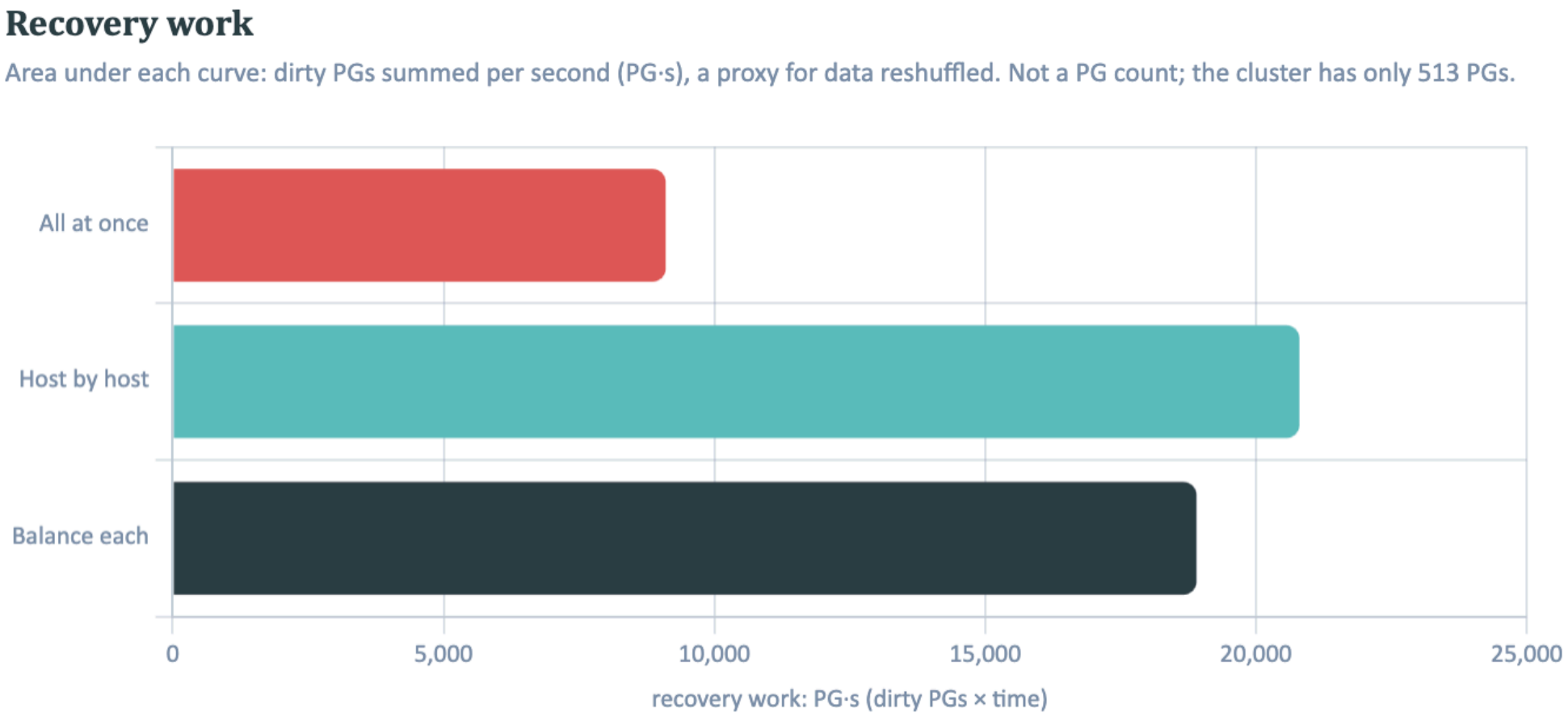


Crushmap changes big-bang vs gradual & +=balancer



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Crushmap changes big-bang vs gradual & +=balancer



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Changelogs

Situation:

- Since September users were reporting seemingly random shell lock-ups from seconds to minutes
- Not reproducible but a real problem.
- This lasted for months where we were in the darks of the problem
- It coincided when I started migrating VMs from VMware/SAN to Proxmox/Ceph a couple of months prior.
- Months later when a pattern started to emerge the question came: when did you start moving VMs?

The result:

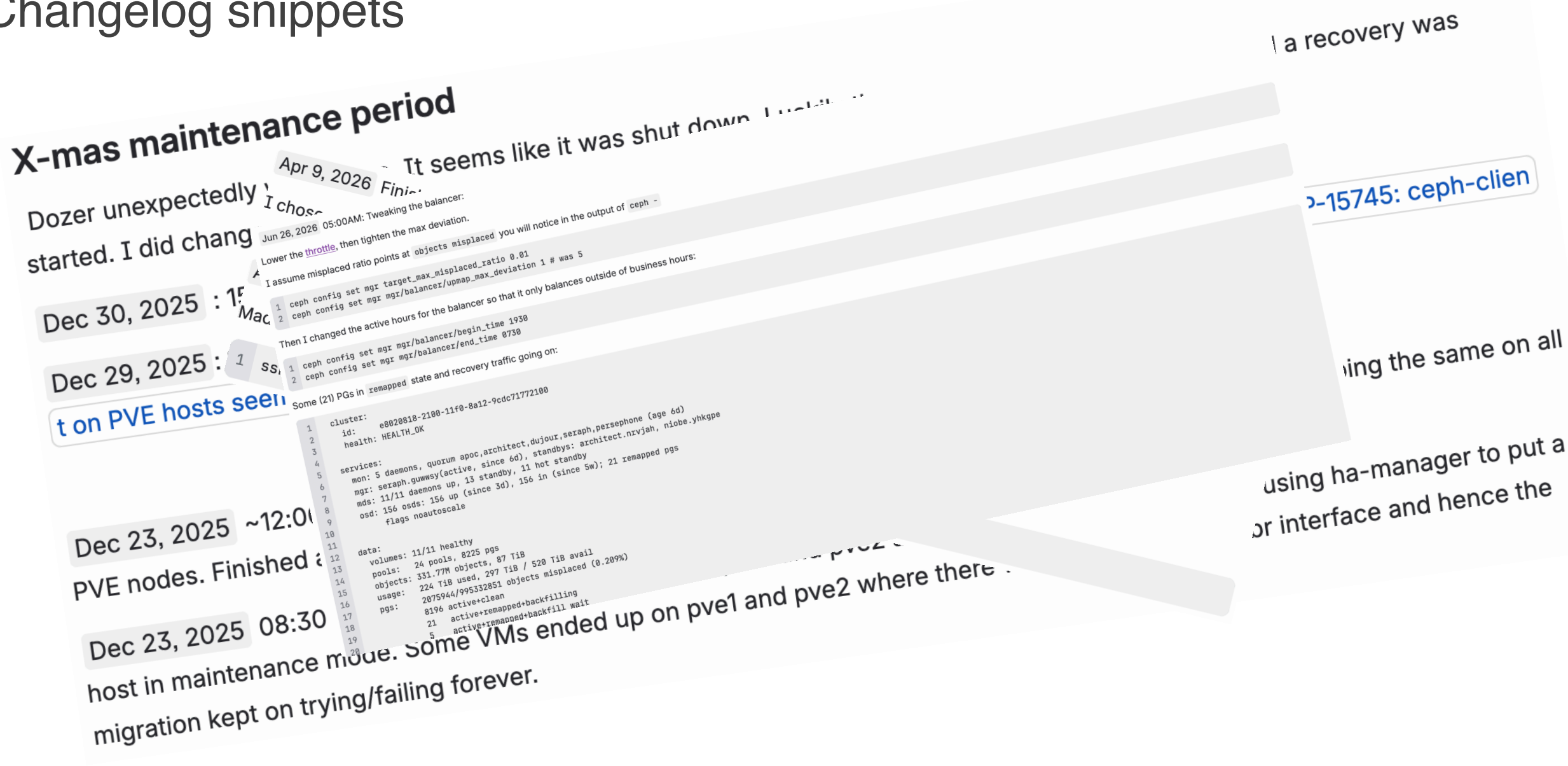
- A months long search digging into Ceph, the network stack and what not.
- After a very long search the problem was related to OpenAFS and volume distribution that got messed up during the migration.

What I wouldn't do twice:

- Not keeping changelogs with accurate timestamps and detailed description of what I changed when.

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Changelog snippets



The Ceph Community & References

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Shout-out to the Ceph community

- People new to Ceph should be aware that there's an active and helpful community behind Ceph.
- A couple of examples:
  - Ceph Slack: Often, you get a reply within minutes.
  - Mentioning a Ceph talk in the mailing list resulted in a recap of the original author in the form of a webinar.
- Big thanks

Some references relevant to my war stories:

- https://www.youtube.com/watch?v=6PQYHlerJ8k (Ceph days NYC 2024 crush / pg / upmap / balancer)
- https://www.youtube.com/watch?v=IWcdftEKby8 (upmap happy hour, part of webinar series)
- https://docs.ceph.com/en/latest/rados/operations/upmap/#enabling
- https://42on.com/how-to-break-your-ceph-cluster/ (point 11 in particular)
- https://www.croit.io/blog/how-not-to-shut-down-a-ceph-cluster (shutting down your Ceph cluster)

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Conclusion

- Ceph is complicated if you're new and it might not behave as you expect it to. Do mess around with a test cluster!
- Please enable the balancer!
- Think about how you approach maintenance windows and think twice about pushing a lot of changes to your cluster.
- Avoid automation pitfalls: one host at a time, aim for `HEALTH_OK` before you continue.
- Old kernels clients might misbehave with CephFS
- MDS: Daemon placement is not carved in stone & MDS requires a LOT of RAM.
- Keep changelogs.
- Don't underestimate the value of the Ceph Community!

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Q&A



BLOCK 1 - Wrap Up:
Group photo,
Coffee Break & Networking

→ See you back at 11.45 a.m.